Undominated Voting in the Clean Immediate-Opt-Out Baseline

A Systemic Rederivation under Simultaneous Ballots

Goal 3 formal workspace

2026-08-04

Contents

1 Status, scope, and provenance

This is the autonomous formal workspace for Goal 3. It does not amend or compile the manuscript. Its purpose is to solve the clean two-round game after replacing the old ballot discipline with **PBE plus local weak-undominance of every vote used at every ballot information set** (PBE-UD).

The input snapshot is the annotated tag **pre-undominated-voting-repair-2026-08-04**, which points to commit **8692a6b31539d117134a580009d4e4dbe783403d**. These artifacts are frozen:

```
formal_model_v6.Rmd
131cc2356cd6318211fdbb9304ac8d7c8a99356837b6e71097011c36ae9c270d

formal_model_v6.pdf
a7d36d5a1fb2d15ba0e40509ad846fbf001b5960232fcbf011d5b32bec298bdf

model_redesign/power_architecture_derivations.Rmd
8fbb7edff59fb0dc6fb36571564ec94d26e66b1211496ed10e9d9191ef2f68c6
```

The post-closure erratum is **quality_reports/2026-08-04_goal2_postclosure_bf_voting_erratum.md**; the autonomous voting contract is **quality_reports/2026-08-04_undominated_voting_protocol_gate0.md**. Gate 0 passed two independent reviews, was reopened after a terminal stress test exposed an overbroad threshold corollary, and passed rereview after repair.

1.1 Result-status convention

Every result is marked **proved**, **checked numerically**, **conjecture**, **pending**, or **rejected**. Enumeration is a regression check, never a substitute for proof.

2 Fixed game contract

There are $N \geq 3$ states: a hegemon H and $m = N - 1$ weak states. Nature draws $\theta \in \{0, 1\}$, observed only by H, with $\mu = \Pr(\theta = 1)$. The primitive domain is

$$0 \leq o_0 < o_1 \leq \bar{y} \leq 1, \qquad \mu \in [0, 1], \qquad \beta \in (0, 1]. \tag{1}$$

The weak institutional pie is one and does not depend on type. Weak outside payoffs are zero. The payoff o_θ is external to that pie. The baseline fixes $\pi_H = 0$ and $b_\theta = 0$ in both rounds. A current agreement that includes H pays it y.

Nature draws the persistent type before R1; only H observes it. In every reached round, a new weak proposer is recognized uniformly and independently, so each weak state has recognition probability $1/m$. Before R1 recognition, the weak states make the collective formation decision described below.

The exogenous rule is $R \in \{U, M\}$, with quotas

$$Q_U = N, \qquad Q_M = q = \left\lfloor \frac{N}{2} \right\rfloor + 1. \tag{2}$$

The proposer is counted as voting yes. All other active players vote simultaneously without observing any other ballot. The complete vote vector and outcome become public only after closure. There is no within-ballot order, no roll call, and no position of H over which to conduct order sensitivity.

A no vote by type θ of H in R1 pays o_θ immediately and irreversibly. Under unanimity that ends the branch. Under majority, a current weak-only agreement passes if weak yes votes reach q; otherwise R2 is a weak-only continuation. If H votes yes but weak votes cause failure, H remains active and R2 follows. R2 is terminal.

The original quota q is not recalculated after opt-out. All R2 flows are multiplied by β in R1 units. If H is active and an R2 ballot fails, type θ receives o_θ at the R2 date and weak states receive zero, regardless of H's R2 vote.

If H opted out in R1, it receives no additional R2 flow. Continuation strategies and beliefs may depend on the complete public post-closure vector, but never on votes before closure.

2.1 Proposals and budgets

When weak state i proposes while H is active,

$$s_i = \left(y, (x_j)_{j \neq i}\right), \qquad y \in [0, \bar{y}], \qquad x_j \geq 0. \tag{3}$$

If H participates, the proposer's residual is

$$x_i^H = 1 - y - \sum_{j \neq i} x_j \geq 0. \tag{4}$$

If H votes no and a majority weak-only agreement passes, y is reabsorbed:

$$x_i^{WO} = 1 - \sum_{j \neq i} x_j. \tag{5}$$

After opt-out, R2 proposals fix $y = 0$. A named offer is paid whenever the current proposal passes, even if its recipient voted no.

2.2 Formation

Before R1 recognition, the m weak states make a collective all-or-nothing formation decision and each pays the external sunk cost χ. For a selected equilibrium under rule R, let T_W^R be expected total gross weak payoff. Under anonymous recognition and symmetric partner selection,

$$V_W^R = \frac{T_W^R}{m}, \qquad \text{formation iff } V_W^R \geq \chi. \tag{6}$$

When asymmetric transfers are allowed, T_W^R/m remains the collective average used by the formation primitive; it is not an individual entry claim for every realized coalition partner.

If the institution does not form, every weak state receives zero and type θ of H receives o_θ.

For any selected collective value v_R, define the formation set

$$\mathcal{F}_R(v_R) = \{\chi \geq 0 : \chi \leq v_R\}.$$

3 Solution concept

Fix a candidate continuation assessment κ. The same κ is used to evaluate both votes against every feasible pure profile of the other voters. Beliefs and continuation strategies are not re-solved separately for each counterfactual profile. They must recursively satisfy PBE-UD.

The weak-vote-passive assessment is a belief discipline: an uninformed weak state's unilateral vote deviation is not treated as a direct signal of θ. A type-contingent action of H may update beliefs after the ballot closes, and Bayes' rule applies whenever possible. This belief discipline neither deletes a ballot action nor selects a vote at indifference.

For an uninformed weak voter j, let $\Omega_{-j}(I)$ contain all feasible pure plans of the other weak voters and all type-contingent plans of H. Other weak actions cannot condition on θ. Define

$$\Delta_j(\omega_{-j} \mid I, \kappa) = \sum_{\theta} \nu_I(\theta) \left[u_j(Y, \omega_{-j}, \theta \mid I, \kappa) - u_j(N, \omega_{-j}, \theta \mid I, \kappa)\right]. \tag{7}$$

Vote Y is weakly dominated by N if $\Delta_j \leq 0$ for every ω_{-j}, with a strict inequality for at least one profile. Reverse inequalities define domination of N. This is one-shot interim Bayesian deletion, not iterative deletion, trembling-hand perfection, or an "as if pivotal" rule.

For H, the comparison is type-specific:

$$\Delta_H(\theta, v_W \mid I, \kappa) = u_H(Y, v_W; \theta, I, \kappa) - u_H(N, v_W; \theta, I, \kappa), \tag{8}$$

where v_W is a counterfactual weak vector, not an observed vector. A PBE-UD is an assessment satisfying: (i) sequential rationality at every information set; (ii) Bayes consistency whenever possible; (iii) the declared weak-vote-passive belief discipline; (iv) ballot strategies supported only on the local $UD_i(I, \kappa)$ correspondence at every on- and off-path ballot; and (v) proposal optimality given the complete voting and continuation assessment. Admissibility does not replace sequential rationality.

3.1 Equality objects

The primary object leaves both votes available when neither dominates. Three selections are then reported:

1. T^Y: yes at every voting indifference;
2. T^N: no at every voting indifference;
3. T^p: an explicitly stated behavioral mixture.

A proposal class is **attained** under a fixed assessment when its stated target payoff is achieved by an available pure proposal and the required equality actions, rather than only approached by a strict sequence. For a cap class, the target already incorporates the fixed marginal-type acceptance probability.

Proposer indifference is separate. Among proposal payoff ties, the legacy selection minimizes H's expected payoff; the opposite selection maximizes it; without selection, the full correspondence remains.

4 Local voting lemmas

Lemma 1 (proved: terminal weak vote). At a reachable terminal ballot, $x_j > 0$ makes N weakly dominated by Y. At $x_j = 0$, the two votes are payoff-equivalent in every feasible profile and both remain admissible.

Proof. Nonpivotal profiles give the same payoff under both actions. In a pivotal profile, Y implements x_j and N produces terminal zero. Every reachable quota admits such a profile. □

Lemma 2 (proved: terminal H vote). With H active in R2,

$$y < o_\theta \Rightarrow UD_H(\theta) = \{N\}, \qquad y > o_\theta \Rightarrow UD_H(\theta) = \{Y\}, \tag{9}$$

while $y = o_\theta$ leaves both votes admissible.

Proof. If H's vote does not change terminal implementation, both actions deliver the same terminal outside payoff. A feasible pivotal weak vector exists; on it, yes gives y and no gives o_θ, with the common R2 discount multiplying both. The sign of $y - o_\theta$ therefore gives the stated correspondence. □

Lemma 3 (proved, conditional scalar reduction). Suppose a weak voter's actions differ only in pivotal profiles; pivotal yes implements x_j; pivotal no always produces the same continuation c_j; and there exists a profile ω_{-j} with positive interim probability of pivotality,

$$\sum_\theta \nu_I(\theta) \mathbf{1}\{j \text{ is pivotal under } (\omega_{-j}, \theta)\} > 0.$$

Then

$$x_j < c_j \Rightarrow UD_j = \{N\}, \qquad x_j > c_j \Rightarrow UD_j = \{Y\}, \tag{10}$$

and equality leaves both votes admissible. This lemma is not used if public vote vectors induce different continuation values.

Proof. For every counterfactual profile, nonpivotal states contribute zero to the payoff difference and pivotal states contribute $x_j - c_j$. Hence $\Delta_j(\omega_{-j}) = p_j(\omega_{-j})(x_j - c_j)$, where $p_j \geq 0$, and the displayed condition supplies a profile with $p_j > 0$. The weak-dominance conclusions follow immediately. □

For an R1 vote by H, let $V_\theta(v_W)$ equal y after current implementation and $\beta C^R_{H,2}(\theta, h_2(Y, v_W))$ after weak-caused failure. Here $h_2(Y, v_W)$ is the complete public post-closure history induced by the current sealed ballot, not information observed during voting.

Lemma 4 (proved: R1 H min–max test). Vote Y is dominated by opt-out exactly when

$$\max_{v_W} V_\theta(v_W) \leq o_\theta \quad \text{and} \quad \min_{v_W} V_\theta(v_W) < o_\theta. \tag{11}$$

The reverse weak inequalities, with at least one strict, characterize domination of opt-out. Otherwise both votes are locally admissible and PBE expected incentives decide their use.

Equivalently, opt-out is dominated by yes exactly when

$$\min_{v_W} V_\theta(v_W) \geq o_\theta \quad \text{and} \quad \max_{v_W} V_\theta(v_W) > o_\theta.$$

Proof. In R1, no gives the constant o_θ against every weak vector, while yes gives $V_\theta(v_W)$. Thus yes is weakly dominated exactly when every value of V_θ is no larger than o_θ and at least one is smaller. Reversing the inequalities gives domination of no. □

An exact offer $y = o_0$ is not automatically rejected: an R2 pooling continuation can pay the low type o_1, and βo_1 may exceed o_0. Under majority with $N = 3$, proposer plus $H = Y$ already attain $q = 2$, so no weak-caused failure vector exists and both H votes remain admissible at $y = o_\theta$.

5 Round 2

Define

$$D = 1 - o_0, \qquad P = 1 - o_1, \qquad L(\nu) = (1-\nu)D, \qquad S(\nu) = \max\{L(\nu), P\}. \tag{12}$$

5.1 Unanimity with H active

Proposition 1 (proved: terminal unanimity correspondence). If $\bar{y} > o_1$, a terminal unanimity PBE-UD exists if and only if at least one class attaining $S(\nu)$ is implemented with probability one by its marginal type of H and every required zero-paid weak voter. Conditional on existence, the recognized proposer obtains $S(\nu)$ and every weak state has ex ante value $S(\nu)/m$.

Under T^Y, low-only at $y = o_0, x_j = 0$ is selected when $L(\nu) > P$, pooling at $y = o_1, x_j = 0$ when $P > L(\nu)$, and both classes are available at equality. The cutoff is

$$\nu_2^* = \frac{o_1 - o_0}{1 - o_0}. \tag{13}$$

Low-only gives H payoffs (o_0, o_1); pooling gives (o_1, o_1). At $L = P$, the proposal tie-break minimizing H selects low-only, the opposite selects pooling, and no selection retains both and their mixtures.

Attainment. If $\bar{y} > o_1$, a best response exists only if a maximizing boundary is accepted with probability one by the marginal type of H and by every required zero-paid weak voter. If the maximizing boundary is not attained, strict offers approach $S(\nu)$ but no pure or mixed proposal attains it.

At $\bar{y} = o_1$, let λ_1 be high-type acceptance at the cap. The cap target gives

$$J(\lambda_1) = \{1 - \nu + \nu\lambda_1\}P. \tag{14}$$

Let e_L indicate that the low boundary is accepted surely by the low type and all required zero-paid weak voters, and let e_J indicate that all required zero-paid weak voters approve the cap surely. Set

$$S_{\text{cap}}(\nu, \lambda_1) = \max\{L(\nu), J(\lambda_1)\}.$$

A cap PBE-UD exists if and only if

$$\{S_{\text{cap}} = 0\} \quad \text{or} \quad \{L = S_{\text{cap}} \text{ and } e_L = 1\} \quad \text{or} \quad \{J = S_{\text{cap}} \text{ and } e_J = 1\}.$$

Conditional on existence, the proposer obtains S_{cap} and every weak state has ex ante value S_{cap}/m. In particular, partial high-type cap acceptance can produce $L \leq J < P = S(\nu)$; the cap value is therefore not generally $S(\nu)$. Existence and value are both assessment-dependent at the cap. When $S_{\text{cap}} = 0$, rejection is the always-attained maximizing class; no boundary acceptance is required. For example, this exception applies at $\nu = 1$ when high H rejects the cap and every feasible agreement gives the proposer zero.

Proof. Terminal single crossing leaves three outcome classes: rejection, low-only, and pooling. Rejection gives the proposer zero. Low-only proposals have supremum $(1-\nu)(1-o_0) = L(\nu)$, attained only at the low boundary with all required equality votes. Pooling proposals have supremum $1 - o_1 = P$, attained only at the pooling boundary with all required equality votes. Any positive weak transfer or excess concession lowers the relevant residual. Thus, with slack capacity, a best response exists exactly when a maximizing boundary attains S; otherwise strict perturbations approach but do not attain it. At the cap, high-type acceptance λ_1 changes the actual pooling target to J, while the low target remains L. The same attainment argument gives the displayed cap test whenever its value is positive. If both active targets are zero, rejection attains the maximum zero. Uniform recognition divides expected total weak surplus equally ex ante. □

5.2 Majority

Let $k = q - 1$, the number of nonproposing weak yes votes required when H does not count.

At a terminal majority proposal, call a set of zero-paid nonproposer votes a **free winning support** when those votes are cast yes with probability one and, together with the proposer's automatic yes, reach q in every type with positive posterior probability. Let z_W be the number of zero-paid weak sure yes votes. When H is active, let $z_H = 1$ if H accepts $y = 0$ surely for every posterior-supported type, and set $z_H = 0$ otherwise. Given $o_0 < o_1$, $z_H = 1$ is possible only at $o_0 = 0, \nu = 0$ under a yes equality action by the low type. After opt-out, set $z_H = 0$.

Proposition 2 (proved). A terminal majority PBE-UD exists if and only if

$$z_W + z_H \geq k.$$

Every such equilibrium gives

$$C^M_{W,2} = \frac{1}{m}, \qquad C^M_{H,2}(\theta) = o_\theta, \tag{15}$$

with proposer payoff one and total weak payoff one. On the regular domain and after R1 opt-out, $z_H = 0$, so the condition reduces to k zero-paid weak sure yes votes. If $o_0 = 0, \nu = 0$ and low H votes yes at equality, only $q - 2$ such weak votes are required; for $N = 3$, none is required. After R1 opt-out, H receives no new R2 flow.

Proof. Paying $\varepsilon > 0$ to any k weak voters forces their yes votes and gives the proposer $1 - k\varepsilon$. Hence a best response must attain one. Attainment requires zero total offers and sure passage in every posterior-supported type. The proposer already supplies one vote; the zero-cost sure votes of H and the weak nonproposers supply exactly $z_H + z_W$. Thus attainment is possible exactly under the displayed support condition. □

Under T^Y this support exists. Under global T^N, or a proper mix that leaves positive probability of missing the quota, no pure or mixed proposer best response exists. Mixing cannot attain a supremum that no pure action in its support attains.

Table 1: Terminal PBE-UD existence and equality sensitivity.

Terminal subgame	Full correspondence / T^Y	Global T^N	Proper mixing at every required equality
U, H active	slack: $S(\nu)$; cap: S_{cap}, each conditional on attainment	no PBE except zero-value cap rejection	no PBE unless one maximizing boundary is accepted surely or the cap value is zero
M, H active	zero-cost passage; proposer one	no PBE	no PBE unless $z_W + z_H \geq k$ surely
M, post-opt-out	weak-only passage; proposer one	no PBE	same sure-support condition

6 Round 1 under majority

This section uses the regular interior domain

$$\mu \in (0,1), \qquad \beta \in (0,1), \qquad 0 < o_0 < o_1 < 1. \tag{16}$$

Write

$$k = q-1, \qquad r = q-2, \qquad c = \frac{\beta}{m}, \tag{17}$$

and define the three relevant proposer limits

$$\begin{aligned} &E = 1 - kc, \\ &B = (1-\mu)\{1 - o_0 - rc\} + \mu c, \\ &C = 1 - o_1 - rc. \end{aligned} \tag{18}$$

Here E is attained exclusion; B is the low-only supremum; and C is the pooling supremum, active only if $\bar{y} > o_1$.

6.1 Voting reduction

Lemma 5 (proved). In every R1 majority subgame whose R2 continuation satisfies Proposition 2,

$$x_j < c \Rightarrow UD_j = \{N\}, \qquad x_j > c \Rightarrow UD_j = \{Y\}, \tag{19}$$

and both votes are admissible at $x_j = c$.

Proof. A pivotal no vote leads either to active-H R2 or post-opt-out weak-only R2. Proposition 2 gives the same discounted weak continuation c in both cases. Lemma 3 applies. □

For $N \geq 4$, a weak-caused failure vector exists after $H = Y$, and R2 majority pays type θ only $\beta o_\theta < o_\theta$. Hence

$$y = o_\theta \implies H_\theta = Y \text{ is weakly dominated.} \tag{20}$$

For $y > o_\theta$, both votes are locally admissible. If p is the equilibrium probability that $H = Y$ implements,

$$H_\theta = Y \quad \iff \quad p \geq \frac{(1-\beta)o_\theta}{y - \beta o_\theta}, \tag{21}$$

with equality permitting mixing. Equation (20) is the dominance restriction; equation (21) is sequential rationality.

6.2 $N \geq 4$

With s c-paid weak supporters:

- $s \leq q-3$: even $H = Y$ cannot implement;
- $s = q-2 = r$: implementation requires $H = Y$;
- $s \geq q-1 = k$: the weak coalition passes without H.

This derives the support size; it does not impose a minimum winning coalition as a primitive.

Theorem 1 (proved: regular majority, $N \geq 4$). Suppose every reached terminal majority subgame has a free winning support, and suppose an R1 exclusion coalition of k voters offered c approves with probability one. A pure or mixed PBE-UD exists if and only if

$$E \geq B \qquad \text{and} \qquad [\bar{y} = o_1 \text{ or } E \geq C]\,. \tag{22}$$

When it exists, every equilibrium is payoff-equivalent to exclusion:

$$T_W^M = 1, \qquad V_W^M = \frac{1}{m}, \qquad u_H^M(\theta) = o_\theta. \tag{23}$$

The recognized proposer obtains E. Multiplicity concerns coalition identity, payoff-irrelevant offers $y < o_0$ to the excluded hegemon, and mixtures over payoff-equivalent exclusion proposals, not the payoff vector. An additional supporter paid c is not payoff redundant: it strictly lowers the proposer residual and therefore cannot appear in an E-attaining best response.

Proof. Exclusion with k offers of c attains E. A strict low-only proposal uses r supporters and approaches

$$(1-\mu)\{1 - o_0 - rc\} + \mu c = B$$

as $y \downarrow o_0$, but cannot attain it because (20) removes low-type yes at $y = o_0$. If $\bar{y} > o_1$, strict pooling with r supporters approaches C as $y \downarrow o_1$, but cannot attain it because high-type yes is removed at the boundary. An oversized separating proposal with k supporters gives

$$E - (1-\mu)y < E$$

and is strictly dominated for the proposer by exclusion with the same support. More supporters only increase cost.

To exhaust mixed weak and hegemonic votes, first note that every weak voter who votes yes with positive probability must receive at least c. Offers strictly above c induce the same sure yes as c and strictly reduce the current residual. Offers at c may mix; conditional on each realized weak vote profile and each type-contingent vote of H, the proposer obtains a convex combination of: continuation, an r-support outcome requiring $H = Y$, and a k-support weak-only outcome. The first two are bounded by the low-only or pooling limits B and C, according to which types of H approve; the last is bounded by E. Randomization by H or by threshold-paid weak voters only convexifies these conditional values and cannot exceed $\max\{E, B, C\}$. Likewise, a proposer mixture can be optimal only if every pure proposal in its support attains the same maximum.

This conditioning is an ex post partition used by the proof. No voter observes or conditions on another current-ballot action before choosing its own vote.

It follows that E is a best response exactly under (22). If either strict supremum exceeds E, no pure action attains the global supremum, and neither voter mixing nor proposer mixing can repair the empty best-response set. When E is maximal, any best response that attains it uses exactly k threshold-paid sure yes votes and excludes H, yielding (23). □

The conditions simplify to

$$E \geq C \iff o_1 \geq c, \tag{24}$$

and

$$E \geq B \iff \begin{cases} \text{always}, & o_0 \geq c, \\ \mu \geq \mu_M := \dfrac{c - o_0}{E - o_0}, & o_0 < c. \end{cases} \tag{25}$$

The parameter condition is the same for $N = 4$ and $N \geq 5$, but the support geometry differs. At $N = 4$, all two nonproposing weak states are needed for exclusion. At $N \geq 5$, there is slack in coalition identity, but not in the total cost kc. Therefore the old claim that large groups always restore existence is **rejected**.

For the required diagnostic $N = 5, q = 3$, including H needs one weak supporter and has candidate boundary cost $o_1 + c$; excluding H needs two weak supporters and costs $2c$. These expressions emerge from (19).

6.3 $N = 3$

Now $m = 2, q = 2, k = 1, r = 0, c = \beta/2$. Proposer plus $H = Y$ always pass, so the exact H thresholds remain admissible. Define

$$\begin{aligned} E &= 1 - c, \\ B &= (1-\mu)(1-o_0) + \mu c, \\ C &= 1 - o_1, \\ Z &= (1-\mu)C + \mu c. \end{aligned} \tag{26}$$

Let λ_0 be low-type acceptance at $y = o_0$, λ_1 high-type acceptance at $y = o_1$, and let $e = 1$ when an exclusion proposal at $x = c$ is supported by a sure weak yes, with $e \in \{0, 1\}$. Exact-threshold payoffs are

$$Q_0(\lambda_0) = (1-\lambda_0)c + \lambda_0 B, \qquad Q_1(\lambda_1) = Z + \lambda_1(C - Z). \tag{27}$$

Theorem 2 (proved: regular majority, $N = 3$). Suppose every reached terminal majority subgame has a valid free winning support.

If $\bar{y} > o_1$, set $V = \max\{E, B, C\}$. A PBE-UD exists under a fixed equality assessment if and only if

$$\{E = V \text{ and } e = 1\} \quad \text{or} \quad \{B = V \text{ and } \lambda_0 = 1\} \quad \text{or} \quad \{C = V \text{ and } \lambda_1 = 1\}. \tag{28}$$

Partial threshold acceptance never exceeds V and cannot rescue an unattained strict maximum. Under T^Y, every class is attained and proposer payoff is $\max\{E, B, C\}$.

If $\bar{y} = o_1$, strict pooling is unavailable. Set

$$V_{\text{cap}} = \max\{E, B, Q_1(\lambda_1)\}. \tag{29}$$

Existence requires an attained maximizer: $E = V_{\text{cap}}$ with $e = 1$, $B = V_{\text{cap}}$ with $\lambda_0 = 1$, or $Q_1(\lambda_1) = V_{\text{cap}}$. The cap class is attained even under partial high-type acceptance and can be uniquely optimal.

Proof. Let x be the offer to the sole nonproposing weak voter. If $x < c$, that voter chooses no, and the proposer maximizes within this region at $x = 0$. If $x = c$, the weak voter may mix; if $x > c$, the voter chooses yes surely, but any excess above c strictly reduces the proposer residual. Thus the only candidate values are exclusion, low-only, pooling, and their exact-threshold mixtures in (27).

Because $c < E$,

$$Q_0(\lambda_0) = (1 - \lambda_0)c + \lambda_0 B \leq \max\{E, B\}.$$

Moreover,

$$Z = B - (1 - \mu)(o_1 - o_0) < B,$$

so $Q_1(\lambda_1) \leq \max\{B, C\}$. Therefore, away from the cap, partial threshold acceptance cannot exceed V; a maximizing class is a best response exactly when its required equality action is actually attained, which gives (28). At the cap, the actual class $Q_1(\lambda_1)$ replaces the unavailable strict-pooling value C and is attained for every fixed λ_1. The same comparison yields (29) and its attainment test. A mixed proposer strategy adds only mixtures over pure actions that already attain the common maximum. □

Table 2: Regular $N = 3$ majority payoff correspondence.

$N = 3$ class	Total weak payoff T_W^M	Collective average $T_W^M/2$	Expected H payoff
Exclusion E	1	1/2	$\bar{o}$
Low-only B	$(1 - \mu)(1 - o_0) + \mu\beta$	total divided by 2	$\bar{o}$
Pooling C	P	$P/2$	o_1
Cap Q_1	$(1 - \mu)P + \mu\{\lambda_1 P + (1 - \lambda_1)\beta\}$	total divided by 2	o_1

At a proposer tie, minimizing H's payoff selects exclusion or low-only over pooling/cap; maximizing H reverses that choice. An $E = B$ tie remains set-valued because both give H $\bar{o}$. Without selection, all attained maximizers and mixtures over them remain.

6.4 Majority equality sensitivity

Table 3: Majority PBE-UD sensitivity to voter equality behavior.

Selection	Terminal requirement	Regular $N \geq 4$	Regular $N = 3$
T^Y	free terminal support exists	Theorem 1 conditions; exclusion	always exists; $\max\{E, B, C\}$
global T^N	terminal best response absent	no full PBE	no full PBE
proper global T^p	terminal support misses quota with positive probability	no full PBE	no full PBE
R1-only mixing with valid R2	threshold supporters must still pass surely	payoff-equivalent exclusion if Theorem 1 holds	Theorem 2 attainment test

7 Round 1 under unanimity

Continue on the regular domain (16). Define

$$\delta = \frac{\beta(m-1)}{m}, \qquad a = 1-\delta, \qquad \nu_2^* = \frac{o_1 - o_0}{1-o_0}. \tag{30}$$

The exact R1 pooling package $y = o_1$ is no longer admissible: after any weak-caused failure, the high type receives only $\beta o_1 < o_1$, whereas current implementation pays exactly o_1. Thus high-type yes is weakly dominated. On a separating path, $H = Y$ reveals the low type, terminal unanimity selects low-only, and $y = o_0$ similarly makes low-type yes weakly dominated. These two observations reject the canonical exact-threshold R1 packages.

7.1 Security floor and weak-PBE completion

When $\bar{y} > o_1$, define

$$K(\mu) = \max\left\{a(1-\mu)D,\ P - \delta D\right\}. \tag{31}$$

The first component is approached by strict low-only proposals $y \downarrow o_0$ that pay each weak voter just above $\beta D/m$. The second is approached by strict pooling proposals $y \downarrow o_1$ with the same punitive weak continuation. At the cap $\bar{y} = o_1$, the second deviation is unavailable and the floor is only $a(1-\mu)D$.

Lemma 6 (proved: unanimity security and completion). Every regular unanimity PBE-UD gives the recognized proposer at least the applicable floor in (31). Conversely, an on-path candidate satisfying Bayes' rule, all vote ICs, local PBE-UD, budget feasibility, and proposer payoff at least $K(\mu)$ can be completed as a PBE-UD under the declared weak-PBE off-path assessment when $\bar{y} > o_1$.

Security proof. Terminal weak continuation never exceeds D/m. For every $\varepsilon > 0$, a strict low-only deviation sets $y = o_0 + \varepsilon$ and pays each weak nonproposer slightly more than $\beta D/m$, forcing all weak yes votes and yielding proposer payoff arbitrarily close to $a(1-\mu)D$. When $\bar{y} > o_1$, the analogous strict pooling deviation sets $y = o_1 + \varepsilon$ and approaches $P - \delta D$. An equilibrium proposal is a best response and therefore cannot pay less than the supremum of these guaranteed deviations. At the cap, the strict pooling deviation is infeasible and that component is absent.

Completion proof. At every globally off-path proposal, assign probability one to the low type both at its ballot and after every $H = Y$, weak-failure history. Select terminal low-only and yes at its required equalities. This gives weak continuation $\beta D/m$. A weak voter offered less votes no; if any such rejection is certain, both types of H opt out. If all weak offers meet the threshold, proposals below o_0 fail, strict low-only is bounded by the first term of K, and strict pooling is bounded by the second. This completes every deviation at no more than K.

This construction uses the declared freedom of weak PBE at globally off-path proposals. It is not robustly claimed for sequential equilibrium, and it does not follow from weak-vote-passive beliefs alone. Together, the two arguments prove the lemma. □

7.2 Exact reduced system

Let h_θ be type θ's probability of voting yes, and let w_j be weak voter j's yes probability. If

$$B_H = (1-\mu)h_0 + \mu h_1 > 0, \qquad \nu = \frac{\mu h_1}{B_H}, \qquad A = \prod_{j\neq i} w_j, \tag{32}$$

then $A > 0$ requires $x_j \geq c(\nu)$ for every nonproposer, where

$$c(\nu) = \frac{\beta S(\nu)}{m}. \tag{33}$$

Write

$$\mathcal{E} = \sum_{j\neq i} x_j - (m-1)c(\nu) \geq 0.$$

The current proposer residual and expected payoff are

$$\begin{aligned} r(\nu) &= 1 - y - \delta S(\nu) - \mathcal{E}, \\ \Pi &= B_H \left[A\, r(\nu) + (1 - A) c(\nu) \right]. \end{aligned} \tag{34}$$

The high-type expected IC is

$$\mathcal{D}_1 = Ay + (1 - A)\beta o_1 - o_1. \tag{35}$$

The low-type difference exceeds the high-type difference:

$$\mathcal{D}_0 - \mathcal{D}_1 \geq (o_1 - o_0)\{1 - (1 - A)\beta\} > 0. \tag{36}$$

Therefore $h_1 > 0 \Rightarrow h_0 = 1$. Reverse separation is impossible.

7.3 Regular correspondence

For $\nu \in (0, \mu]$, define

$$B(\nu) = \frac{1 - \mu}{1 - \nu}, \qquad h_1(\nu) = \frac{\nu(1 - \mu)}{\mu(1 - \nu)}. \tag{37}$$

Theorem 3 (proved: regular unanimity PBE-UD). Conditional on a terminal equality assessment that attains the R2 value, a regular unanimity PBE-UD exists if and only if

$$\boxed{\bar{y} > o_1 \quad \text{and} \quad \mu > \nu_2^*} \tag{38}$$

or, equivalently, $P > L(\mu)$.

Every equilibrium candidate has $\nu \in (\nu_2^*, \mu]$, so $S(\nu) = P$, $h_0 = 1$, and the tuple $(\nu, A, y, (x_j, w_j)_{j \neq i})$ satisfies:

$$\begin{aligned} &0 < A \leq 1, \qquad o_1 < y \leq \bar{y}, \qquad x_j \geq \frac{\beta P}{m} \text{ if } w_j > 0, \\ &x_j > \frac{\beta P}{m} \Rightarrow w_j = 1, \qquad r(\nu) \geq 0, \qquad \Pi \geq K(\mu). \end{aligned} \tag{39}$$

If $\nu < \mu$, the high type mixes, $h_1 = h_1(\nu) \in (0, 1)$, $A \in (0, 1)$, and $\mathcal{D}_1 = 0$. If $\nu = \mu$, both types vote yes and $\mathcal{D}_1 \geq 0$; A may lie anywhere in $(0, 1]$. At least one weak voter who mixes must receive exactly $\beta P/m$. For $N = 3$, there is only one such weak nonproposer, so $A < 1$ implies $\mathcal{E} = 0$.

Proof. Low-only with $h_0 = 1, h_1 = 0$ has maximum

$$\Pi_L(A) = (1 - \mu) \left[aD - (1 - A)(1 - \beta) \right]. \tag{40}$$

For $A < 1$ this is strictly below the first component of K; for $A = 1$, attaining the boundary would require the dominated action $H_0 = Y$ at $y = o_0$. Delay yields only $(1 - \mu)\beta D/m < a(1 - \mu)D$. Thus any equilibrium needs $h_1 > 0$.

For every candidate with $\Pi \geq K$, it must be that $S(\nu) = P$. If $S(\nu) = L(\nu)$, the maximum payoff inside the brackets in (34) is strictly below $aL(\nu)$, while $B(\nu)aL(\nu) = a(1 - \mu)D$, contradicting $\Pi \geq K$. Therefore $\nu > \nu_2^*$, which requires $\mu > \nu_2^*$. The high type's positive participation requires $y > o_1$, which requires $\bar{y} > o_1$.

For sufficiency, set $\nu = \mu, A = 1, \mathcal{E} = 0$, and choose

$$y = o_1 + \eta, \qquad 0 < \eta \leq \min\{\bar{y} - o_1,\ aP - K(\mu)\}. \tag{41}$$

When $P > L(\mu)$, both entries in K are strictly below aP. The proposal gives $\Pi = aP - \eta \geq K$, and Lemma 6 supplies the global completion. □

The theorem includes three on-path forms:

1. **pure pooling with all weak yes:** $\nu = \mu, A = 1$;
2. **pure H pooling with weak mixing:** $\nu = \mu, A \in (0, 1)$;
3. **semi-pooling:** $\nu \in (\nu_2^*, \mu)$, with high-type H mixing.

For high-type mixing, its binding IC gives

$$y_1(A) = \beta o_1 + \frac{(1-\beta)o_1}{A} > o_1 \tag{42}$$

and, with minimum weak transfers,

$$\Pi_{SP}(\nu, A) = B(\nu)\,[P - \delta P - (1-A)\{1 - \beta(o_1 + P)\}]\,. \tag{43}$$

The feasibility tests are $y_1(A) \leq \bar{y}$, $1 - y_1(A) - \delta P \geq 0$, and $\Pi_{SP} \geq K$. Equation (43) applies when the high type is indifferent, $\mathcal{D}_1 = 0$, including the $\nu = \mu$ endpoint when equality holds. If $\nu = \mu$ and $\mathcal{D}_1 > 0$, the candidate must instead be evaluated directly from (34); relative to the binding formula its payoff is reduced by $A\{y - y_1(A)\}$ and by $A\mathcal{E}$.

7.4 Payoffs, mixing, and selections

For every regular candidate,

$$T_W^U = B(\nu)\,[A(1-y) + (1-A)\beta P] < P, \tag{44}$$

and $V_W^U = T_W^U/m$ under the collective formation convention.

Let $v_H = Ay + (1-A)\beta o_1$. High-type mixing sets $v_H = o_1$; pure high acceptance requires $v_H \geq o_1$. Since the low type accepts surely,

$$E[u_H^U] \geq o_1. \tag{45}$$

For the T^Y construction (41),

$$\Pi = aP - \eta, \qquad T_W^U = P - \eta, \qquad E[u_H^U] = o_1 + \eta. \tag{46}$$

A proposer mixture adds no local class: every pure proposal in its support must satisfy (39) and deliver the same proposer payoff. Such mixtures convexify supported outcome vectors.

Under global T^Y, the pure construction is available. Under global T^N, terminal R2 has no best response, so the complete game has no PBE. A proper global T^p likewise fails unless a maximizing terminal boundary is still accepted with probability one. With a valid T^Y terminal continuation but mixing only in R1, the full correspondence in Theorem 3 applies.

The proposer tie-break operates only among actual payoff ties within a fixed completion. It does not rank equilibria sustained by different off-path belief systems. Within a tied support, minimizing H selects the smallest expected value in (45); maximizing H selects the largest; without selection, all tied candidates remain.

Table 4: Regular R1 unanimity survival under PBE-UD.

Former R1-U class	PBE-UD status on the regular domain	Reason
exact low-only $y = o_0$	rejected	low-type yes is dominated on the separating path
exact pooling $y = o_1$	rejected	high-type yes is dominated
deliberate rejection/delay	rejected	proposer payoff lies below the security floor
strict overpaid pooling	survives with a new proof	available iff $\bar{y} > o_1$ and $\mu > \nu_2^*$
semi-pooling / weak mixing	newly required	part of the full correspondence

8 Boundaries and one-sided limits

These are literal boundary games unless explicitly labeled as limits. Equality behavior is part of each statement; no result is filled in by silent continuity. Throughout the literal boundary subsections the prior remains interior, $\mu \in (0, 1)$; prior endpoints are treated only as one-sided limits below.

8.1 $o_0 = 0$, $\beta < 1$

At $y = o_0 = 0$, low-type implementation, continuation, and opt-out all pay zero. Low-type yes therefore remains admissible. Let

$$K_0 = \begin{cases} \max\{a(1-\mu), P - \delta\}, & \bar{y} > o_1, \\ a(1-\mu), & \bar{y} = o_1, \end{cases} \tag{47}$$

so the strict-pooling component is absent, rather than set to zero, at the cap.

Proposition 3 (proved, T^Y). Exact low-only under unanimity gives proposer payoff $a(1-\mu)$. If $\bar{y} > o_1$, it exists if and only if $a(1-\mu) \geq P - \delta$; if $\bar{y} = o_1$, it always exists. Strict high-participation equilibria also exist when $\bar{y} > o_1$ and $\mu > o_1$, with Theorem 3's candidate system evaluated at $D = 1$. Thus at least one unanimity PBE-UD exists everywhere on this boundary under T^Y, though the correspondence can be multiple.

Proof. At $o_0 = 0$, low-type current implementation, opt-out, and the selected low terminal continuation all pay zero, so the low type's exact yes vote is admissible. With all required equality votes selected yes, the exact low-only proposal pays the proposer $a(1-\mu)$. Under slack capacity its only active security competitor is the strict-pooling limit $P - \delta$, giving the stated comparison; at the cap that deviation is unavailable. If the exact class loses under slack capacity, then $a(1-\mu) < a - o_1$, so $\mu > o_1/a > o_1$; the strict high-participation condition holds. Substituting $D = 1$ in Theorem 3 gives the remaining candidates. □

For a general fixed equality assessment, the exact low-only class attains $a(1-\mu)$ only when the low type of H and every equality-paid weak voter needed for unanimity vote yes with probability one. It is a proposer best response only if this attained value is no smaller than the active strict-pooling security component $P - \delta$ when $\bar{y} > o_1$. If the required equality actions do not attain the class, strict deviations approach its full value and the partially implemented class cannot be a best response. The strict high-participation candidates are governed by the full system in Theorem 3 with $D = 1$, rather than by an equality convention.

Majority boundary classification (proved). Under global T^Y, exact low-only is also attained. For $N \geq 4$, define

$$B_0 = (1-\mu)(1-rc) + \mu c. \tag{48}$$

Let $V_0 = \max\{E, B_0\}$. Under a fixed equality assessment, a full $N \geq 4$ majority PBE-UD exists if and only if: terminal free winning supports are valid; at least one class equal to V_0 is attained; and, when $\bar{y} > o_1$, $V_0 \geq C$. Attaining E requires k threshold-paid sure weak yes votes; attaining B_0 requires the low type of H and the r

threshold-paid weak supporters to vote yes surely. Under global T^Y, both classes are attained. For $N = 3$, apply Theorem 2 with B_0 in place of B: use $\max\{E, B_0, C\}$ off the cap and $\max\{E, B_0, Q_1(\lambda_1)\}$ at the cap, always conditional on valid terminal support and the stated attainment test.

Proof. Setting $o_0 = 0$ makes the low threshold an admissible equality and replaces the regular low supremum B by the attained value B_0. The support enumeration in Theorems 1 and 2 is otherwise unchanged. The strict pooling target remains nonattained off the cap, while Proposition 2's corrected free-support condition governs every reached terminal majority subgame, including a degenerate low posterior. Comparing the attained classes with C, or with the actual cap class Q_1, yields the stated tests. □

8.2 $o_1 = 1$, $\beta < 1$

Boundary classification (proved).

Here $\bar{y} = o_1 = 1$, so strict high participation is impossible. If $o_0 > 0$, regular-style low-only remains unattained and unanimity has no PBE-UD. If $o_0 = 0$, exact low-only exists exactly when its low-type and weak equality actions attain the class; in particular it exists under T^Y.

Conditional on a valid terminal support, majority pooling is payoff-irrelevant. At $N = 3$, proposer value is $\max\{E, B\}$ provided a maximizing class is attained as in Theorem 2. At $N \geq 4$ and $0 < o_0 < 1$, existence requires $E \geq B$ and an attained exclusion support, and the outcome is exclusion. The corner $o_0 = 0, o_1 = 1$ is governed by the preceding subsection, where exact low-only may also attain the maximum.

Proof. Capacity rules out every offer strictly above the high threshold. When $o_0 > 0$, exact low yes is removed by the regular weak-failure test and strict low offers have an unattained supremum, so unanimity has no proposer best response. At $o_0 = 0$, Proposition 3 applies. Under majority, the pooling residual is zero and cannot exceed exclusion or low-only. Theorem 2 then gives the $N = 3$ attainment test, while Theorem 1 gives exclusion for $N \geq 4$ on the stated noncorner domain. □

8.3 $\beta = 1$

Boundary classification (proved conditional on the stated equality assessment).

At no discounting, a failure continuation can pay each type exactly its outside option. Exact H thresholds again remain admissible, but a class that relies on equality votes is available only when those votes actually attain it. The formulas below first identify the candidate values and then separate the full attainment statement from the global T^Y selection.

For unanimity, set

$$K_1 = \max\left\{\frac{L(\mu)}{m},\ P - \frac{m-1}{m}D\right\}. \tag{49}$$

Under global T^Y, if $L(\mu) > P$, exact low-only and deliberate continuation give proposer payoff $L(\mu)/m$. If $P > L(\mu)$, exact pooling and deliberate continuation give P/m. At $L(\mu) = P$, exact low-only, exact pooling, and deliberate continuation all enter the unselected correspondence when their required equality actions are attained. Exact low-only also survives in the second region when

$$P - \frac{m-1}{m}D \leq \frac{L(\mu)}{m}. \tag{50}$$

For a general fixed equality assessment, every exact low-only, exact pooling, or delay class must be attained by all equality actions it requires; the proposer can mix only among attained pure classes that share the maximum. Strictly overpaid pooling is evaluated from the general system (34). A payoff strictly above K_1 is sufficient; equality with K_1 can also remain in the unselected correspondence if it attains the common maximum. Under a minimum-H proposer tie-break, such an equality candidate survives only if it also minimizes H's payoff among tied best responses.

For majority, use $c = 1/m$ and write $D_O = E - (1-\mu)o_0$ for the oversized separating value. Under global T^Y, the exact attained proposer value, both with slack capacity and at the cap, is

$$\max\{E,\ (1-\mu)(1-o_0-rc)+\mu c,\ D_O,\ 1-o_1-rc\}, \tag{51}$$

over feasible classes. The third entry is redundant for value comparisons because $D_O \le E$, but it records the oversized class explicitly. Delay gives $c \le E$, with equality only for $N = 3, 4$.

With slack capacity, a general fixed equality assessment uses the same four candidate suprema in (51), together with delay c. A full PBE-UD exists if and only if terminal free winning supports are valid and at least one class equal to the largest feasible candidate is actually attained. Thus a unique low-only or pooling maximum requires its exact boundary actions to implement surely; otherwise strict perturbations approach the candidate but no best response exists. An attained exclusion class, or delay when $E = c$, can support a tie at the maximum.

At $\bar{y} = o_1$, the pooling entry $C = 1 - o_1 - rc$ must instead be replaced for **every** N, not only $N = 3$, by the actual cap correspondence. For a cap proposal with exactly r potentially approving weak supporters, let $R_s = 1 - o_1 - \sum_{j\in s} x_j$ be the current proposer residual, let A_s be the probability that all those supporters vote yes, and let $\lambda_{1,s}$ be high-type acceptance at that proposal. Each supporter has $x_j \ge c$; a strict offer forces yes, while $x_j = c$ permits mixing. Low H votes yes surely because $o_1 > o_0$, whereas high H may mix at equality. The actual payoff is

$$Q_s(\kappa) = c + A_s\{1-\mu+\mu\lambda_{1,s}\}(R_s - c).$$

For the canonical threshold package $x_j = c$ for all r supporters, $R_s = C$, and hence

$$Q_{1,N}(A,\lambda_1) = c + A(1-\mu+\mu\lambda_1)(C-c).$$

When the threshold supporters approve surely, this reduces to

$$Q_{1,N}(1,\lambda_1) = (1-\mu)C + \mu\{\lambda_1 C + (1-\lambda_1)c\}.$$

For $N = 3$, $r = 0$ and $A = 1$, so this is exactly $Q_1(\lambda_1)$ in (27). Let $\overline{Q}_\kappa$ be the supremum of $Q_s(\kappa)$ over all feasible r-support cap proposals under the fixed, proposal-contingent assessment, set it to $-\infty$ if that set is empty, and let $e_Q = 1$ exactly when that supremum is attained. Proposals with fewer than r potential supporters yield continuation c; proposals with at least $k = r + 1$ potential supporters are bounded by the exclusion target E. Therefore the cap target is

$$V^\kappa_{\text{cap}} = \max\{E, B, \overline{Q}_\kappa\},$$

over feasible components. A full cap PBE-UD exists if and only if terminal free winning supports are valid and a component equal to V^κ_{cap} is attained: exclusion (or delay when $E = c$), the exact low boundary for B, or a cap proposal with $e_Q = 1$. Proposer mixing is only over attained pure maximizers. Under global T^Y, $A = \lambda_1 = 1$, so $\overline{Q}_\kappa = C$, all feasible exact classes are attained, and (51) follows.

The cap distinction is substantive. For $N = 4,\ o_0 = 1/10,\ o_1 = \bar{y} = 1/5,\ \mu = 1/2$, take $A = 1$ and $\lambda_1 = 9/10$. Then

$$E = \frac{1}{3}, \qquad B = \frac{9}{20}, \qquad C = \frac{7}{15}, \qquad Q_{1,4} = \frac{23}{50}.$$

Thus $C > Q_{1,4} > B > E$. The full-acceptance value C is unavailable at the cap, but the partial cap class is attained and can be the unique proposer best response under a fixed assessment that selects high-type no at the other cap proposals. This is a PBE-UD possibility, not an $N = 3$ exception.

A global T^N still has no full equilibrium because terminal majority lacks a proposer best response; any R1-no sensitivity with a valid terminal T^Y must be labeled hybrid.

Proof. With $\beta = 1$, implementation and a failure continuation can coincide exactly with each type's outside payoff, so exact H-threshold votes re-enter the local undominated correspondence. Under unanimity, direct enumeration of low-only, pooling, and delay gives the two entries of K_1; overpaid candidates continue to use (34). Under majority, the support enumeration from Theorems 1 and 2, now allowing exact thresholds, gives the four displayed current-proposal values; deliberate delay adds c. With slack capacity, strict perturbations retain those suprema and the usual attainment test.

At the cap, exactly r weak supporters pass only when $H = Y$ and all supporters approve. Failure gives the proposer continuation c. Low H accepts surely, and high H's equality action has probability $\lambda_{1,s}$, so direct expectation yields $Q_s(\kappa)$. Fewer supporters cannot pass; at least k can pass without H and cannot beat E. This exhausts the cap support geometry. Taking the supremum over the proposal-contingent assessment and requiring its attainment proves the cap test. Only ties among attained pure maxima can mix. □

8.4 Priors as one-sided limits

One-sided-limit classification (proved).

Literal $\mu = 0, 1$ games require a convention for zero-probability types. The objects here are cluster limits from $\mu \in (0, 1)$.

For regular unanimity, $\nu_2^* > 0$, so no equilibrium sequence reaches $\mu \to 0^+$. As $\mu \to 1^-$, pure-pooling limits exist when $\bar{y} > o_1$. For any fixed

$$0 < \eta \leq \min\left\{\bar{y} - o_1,\ aP - \max(0, P - \delta D)\right\},$$

the limiting vector is

$$\left(aP - \eta,\ P - \eta,\ \frac{P - \eta}{m},\ o_1 + \eta\right). \tag{52}$$

Semi-pooling can add cluster points; no uniqueness claim is made.

For $N \geq 4$, regular majority exclusion has a lower-endpoint sequence only if $o_0 \geq c$ and, when $\bar{y} > o_1$, $o_1 \geq c$. An upper-endpoint sequence requires only the latter capacity-dependent pooling condition. $N = 3$ under T^Y has the direct limits of $\max\{E, B, C\}$. These are limits of regular equilibria, not solutions of new degenerate games.

Proof. Take $\mu \to 0^+$ and $\mu \to 1^-$ in the regular existence conditions and payoff formulas while holding the remaining primitives fixed. The strict lower cutoff excludes a unanimity sequence at zero; substituting the pure construction gives (52). Equations (24) and (25) give the majority endpoint conditions. □

9 Entry and institutional comparison

Let

$$\bar{o} = (1 - \mu)o_0 + \mu o_1. \tag{53}$$

The regular common domain consists of (38), a valid terminal equality support, and the applicable majority existence condition: Theorem 1 for $N \geq 4$, or the attainment condition in Theorem 2 for $N = 3$. Because regular unanimity requires $\bar{y} > o_1$, the $N = 3$ cap class Q_1 lies outside this common regular domain.

Proposition 4 (proved: regular entry nesting). In every regular unanimity PBE-UD,

$$T_W^U < P. \tag{54}$$

In every regular majority PBE-UD on the common domain,

$$T_W^M \geq P. \tag{55}$$

Therefore

$$V_W^U < V_W^M, \qquad \mathcal{F}_U \subsetneq \mathcal{F}_M. \tag{56}$$

Proof. Equation (44) is a probability-weighted combination of $1-y < P$ and $\beta P < P$, multiplied by $B(\nu) \leq 1$. For $N \geq 4$, majority exclusion gives total one. For $N = 3$, an attained exclusion maximizer has $E \geq C = P$; an attained low-only maximizer has $B \geq C = P$, and its total weak payoff is $B + \mu c \geq P$; pooling gives exactly P. Mixtures preserve the inequality. Since the unanimity inequality is strict, the interval $(V_W^U, V_W^M]$ is nonempty and establishes the proper formation-set inclusion in (56). □

For a selected pair $(v_U, v_M) = (V_W^U, V_W^M)$:

1. both institutions form if $\chi \leq v_U$;
2. only majority forms if $v_U < \chi \leq v_M$;
3. neither forms if $\chi > v_M$.

There is no regular common-domain region in which only unanimity forms. Unlike the old scalar result, v_U is generally a correspondence because strict pooling, weak mixing, and semi-pooling can coexist.

Proposition 5 (proved: H comparison). In every regular majority PBE-UD,

$$\bar{o} \leq E[u_H^M] \leq o_1. \tag{57}$$

In every regular unanimity PBE-UD,

$$E[u_H^U] \geq o_1. \tag{58}$$

Hence unanimity weakly benefits H whenever both rules form on the common domain. Equality is possible only in the $N = 3$ majority pooling class when unanimity keeps the high type exactly indifferent; the T^Y strict-pooling construction gives a strict advantage $o_1 + \eta > o_1$.

Proof. For regular $N \geq 4$, Theorem 1 gives H the type-specific outside payoff, whose expectation is $\bar{o}$. For $N = 3$, exclusion and low-only also give $\bar{o}$, whereas pooling and the cap class give o_1; mixtures remain between these values. Under unanimity, high-type sequential rationality requires its yes payoff to be at least o_1. Single crossing makes the low type accept surely and, on the surviving pooling continuation, gives it the same or a larger payoff. Hence the expected unanimity payoff is at least o_1. The equality and strict cases follow from the identified classes and (46). □

The mechanism therefore survives, but with new mathematics: exact pooling at o_1 is removed, strict overpayment or mixing replaces it, and majority exclusion becomes the unique regular payoff for $N \geq 4$ whenever it exists.

9.1 Comparative statics

- Unanimity existence is monotone in the prior: it begins strictly above ν_2^*, and it requires positive capacity slack $\bar{y} - o_1$.
- Under the T^Y pure construction, greater η transfers surplus one-for-one from weak states to H; the admissible upper bound is $\min\{\bar{y} - o_1, aP - K\}$.
- For majority with $N \geq 4$, if $o_0 < c$, increasing μ past μ_M removes the low-only nonattainment threat. If $o_0 \geq c$, that threat never binds. When capacity is slack, $o_1 \geq c$ removes the pooling threat.
- Increasing β raises the weak approval price $c = \beta/m$ and can shrink majority's existence domain. Boundary $\beta = 1$ is not obtained by simply substituting into the regular theorem because exact H thresholds re-enter.

10 Reproducible verification

All computation is in new, separate R scripts:

- **verify_undominated_voting_gate0.R**: finite ballot checks of the local dominance lemmas, the $N = 3$ exception, and the pooling-continuation counterexample, plus the zero-value terminal rejection exception;
- **verify_undominated_voting_regular.R**: majority support enumeration, direct low-only and pooling branch assertions, regular existence identities, $N = 5, q = 3$, and three unanimity witnesses (strict pooling, pure-H weak mixing, and semi-pooling);
- **verify_undominated_voting_boundaries.R**: $o_0 = 0$, $o_1 = 1$, $\beta = 1$, the all-N cap-mixing formula and $N = 4$ counterexample, and one-sided-limit algebra, with every row labeled by its equality assessment where applicable;
- **verify_undominated_voting_all.R**: orchestration wrapper.

The current generated checks report:

Table 5: Reproducible Goal 3 verification summary.

Verifier	Result	Evidentiary status
Gate 0 finite ballots	127/127 PASS	checked numerically
Regular-domain identities and enumeration	10085/10085 PASS	checked numerically
Boundary identities	2881/2881 PASS	checked numerically

Every CSV output under **tables/** records source, date, and status. The regular and boundary region files carry parameter columns directly; the Gate 0 finite-ballot file records each tested value in its check identifier and detail field. These checks verify implementation of the displayed identities; the general results rest on the analytical proofs above.

11 Result ledger

Table 6: Goal 3 formal result ledger.

Object	Final status	Goal 3 conclusion
Simultaneous sealed ballot	proved	preserved; no order of H exists
Terminal U correspondence	proved	slack value S; cap value S_{cap}, each conditional on attainment
Terminal M payoff	proved	unique conditional on endogenous free support, including the degenerate free-H exception
Exact R1-U low-only	rejected	dominated H_0-yes on separating path
Exact R1-U pooling	rejected	dominated H_1-yes
Strict/semi-pooling under U	proved	full correspondence in Theorem 3
Old regular U existence condition	rejected	replaced by $\bar{y} > o_1, \mu > \nu_2^*$
Old majority floor F_M	rejected	replaced by attained E and nonattained B, C
Old large-N interval $[F_M, 1]$	rejected	regular $N \geq 4$ payoff collapses to exclusion
Old No-Cheap-H condition	rejected	replaced by (25) and (24)
Entry nesting	proved anew	strict on the regular common domain
Informational advantage for H	proved anew	$E[u_H^U] \geq o_1 \geq E[u_H^M]$

Object	Final status	Goal 3 conclusion
Global T^N equilibrium	rejected generically	terminal proposer best response absent, except for the zero-value terminal-U cap rejection corner

12 Scope and limitations

The regular unanimity sufficiency proof is a weak-PBE construction. It uses off-path proposal beliefs that are not linked by consistency to nearby on-path proposals. No claim is made for sequential equilibrium, trembling hand perfection, coalition-proofness, endogenous rule choice, or $\pi_H > 0$. Equality selections are reported rather than silently fixed.

Most importantly, none of these results uses public or sequential voting inside a ballot. Later voters never observe earlier votes because there are no later voters within the simultaneous ballot. Baron–Ferejohn motivates excluding weakly dominated voting actions; it does not supply this paper's protocol.